

Copyright © 2026 by George M. Van Treeck. Creative Commons 4.0.

Some set and number properties underlying geometry and physics

George M. Van Treeck 

(Dated: September 15, 2026)

Abstract

Euclidean volume and some distance equations are proved to be derived from ordered combinations (n-tuples) of elements from a list of countable sets. The domain sets and volume domain values are both commutative and linearly sequenced, which is proved to be a cyclic order limited to at most 3 elements: for example, a set of 3 distance dimensions, [width, height, depth], and a set of 3 non-distance dimensions, [time, mass, charge]. Dividing the dimensions of Euclidean distance, time, mass, and charge into constant-sized intervals, for i in 1 to n , there are 3 direct and 3 inverse proportion ratios of the i -th constant Euclidean distance interval length to the i -th constant interval lengths of time, mass, and charge. Rigorous derivations of 18 well-known gravity, charge, electromagnetism, relativity, quantum physics equations and 10 related constants (including the gravity and Planck constants) from the math proofs and ratios are presented. The need for dimensional analysis is eliminated. Quantum extensions to the Schwarzschild metric, Newton's gravity force, and Coulomb's charge force predict that, for both mass and charge, the forces peak at the Planck unit length and then go to 0 as the distance between point-like particles goes to 0.

CONTENTS

Popular summary	4
Introduction	4
Insufficient reduction	4
First reduction step	5
Second reduction step	6
Third reduction step	7
Volume	7
Euclidean volume	7
Distance	8
Dot product	8
Minkowski distance (p-norm)	9
Distance inequalities	9

Metric Space	10
Properties limiting a set to at most 3 elements	11
Derivation of physics equations and constants	13
3 direct and 3 inverse proportion ratios	13
G , Newton, Gauss, Poisson gravity laws	14
Special relativity	14
Einstein's general relativity	15
Analytic general relativity solutions	15
Schwarzschild's time dilation and metric	16
k_e , ε_0 , E , Coulomb and Gauss laws	17
B , μ_0 , and Lorentz's law	18
Faraday's law	19
Maxwell's wave equations	20
Classic electron radius, r_e	20
Fine structure constants, α and α_G	21
Rydberg constant, R_∞	21
$\hbar$, h , and Planck-Einstein relation	21
Values of the unit ratios and Planck units	22
Compton wavelength, λ	23
Schrödinger's wave equation	23
Dirac's wave equation	24
Total of a type	24
Quantum extension to Schwarzschild's metric	24
Quantum extension to Newton's gravity force	25
Quantum extension to Coulomb's charge force	25
Insights and implications	26
Data Availability Statement	32
References	33

Keywords: mathematical physics, gravity, charge, electromagnetism, relativity, quantum.

POPULAR SUMMARY

Simpler, more rigorous derivations of 18 well-known gravity, charge, electromagnetism, relativity, quantum physics equations and 10 related constants (including the gravity and Planck constants) from some math proofs and ratios are presented. The math proofs and ratios are a framework for teaching physics. Quantum extensions to the Schwarzschild metric, Newton's gravity force, and Coulomb's charge force predict that, for both mass and charge, the forces peak at the Planck unit length and then go to 0 as the distance goes to 0.

INTRODUCTION

Insufficient reduction

The philosophy of reductionism posits that complex phenomena can be reduced to simpler components and relations. But, most physics equations have lacked derivations from a few set and number properties, for example, Newton's gravity, Coulomb's charge, Maxwell's wave equations, Einstein's field equations (general theory of relativity), and Schrödinger's wave equation (quantum physics) [1–5].

Classical, relativity, and most quantum theories assume only 3 dimensions of space [1–5]. But, integration, Hilbert spaces, linear algebra, Riemann manifolds, pseudo-Riemann manifolds, and so on, have no restriction on the number of dimensions of space [4, 6–9].

Further, many physics equations contain volume or distance. But mathematical analysis (Riemann and Lebesgue integrals, Lebesgue and other measures) defines Euclidean volume as the product of interval lengths. And analysis defines Euclidean distance, dot product, inner product, and metric space [6, 7]. Justifications for the definitions use rigorous finger-pointing to diagrams, Euclidean geometry, Cartesian geometry, and “It's an abstraction.”

The first reduction step, in this article, proves that some well-known volume and distance equations are reduced to (derived from) a few simple set and number properties. The second step identifies three properties of the sets and corresponding volume and distance equations that are salient to physics. The third reduction step shows that some well-known physics equations, constants, and assumptions are reduced to (derived from) two consequences of those three properties.

First reduction step

If the underlying principles are simple, then the math proofs should also be simple, trivial even. But even apparently trivial proofs can have subtle flaws. Therefore, each math proof, in this article, has a corresponding formal proof, which has been verified using the Rocq proof verification system (formerly known as Coq) [10], a verification tool used by mathematicians world-wide. See the data availability section for more details.

Let $|x_i|$ be the cardinal of (number of elements in) the abstract, countable set, $x_i \in \{x_1, \dots, x_n\}$. And let the value, v_c , be the number of ordered combinations (n-tuples) of the elements of $x_1, \dots, x_n$. Notionally:

$$\forall x_i \in \{x_1, \dots, x_n\}, \quad \bigcap_{i=1}^n x_i = \emptyset, \quad |x_i| \in \{0, \mathbb{N}\}, \quad v_c = \prod_{i=1}^n |x_i|. \quad (1)$$

The Euclidean volume equation will be proved to be the trivial case where each n-tuple corresponds to a scalar value, $\kappa \in \mathbb{R}$ and v_c is the total number of n-tuples:

$$\forall n, v_c \in \mathbb{N}, \kappa \in \mathbb{R}: \quad v_c \kappa = \left(\prod_{i=1}^n |x_i|\right) \kappa \quad \Rightarrow \quad \exists v, s_i \in \mathbb{R}: \quad v = \prod_{i=1}^n s_i. \quad (2)$$

Note that the goal, here, is to derive the Euclidean volume equation and is *not* a goal to prove that each s_i is an interval length and *not* a goal to prove that $\prod_{i=1}^n s_i \subseteq \mathbb{R}^n$.

For all $n > 1$, there are cases where different domain values, $|x_1|, \dots, |x_n|$, multiplied yield the same range value, v_c . Inferring a single, countable domain value, d_c , from the countable range value, v_c , requires an inverse (bijective) function, $d_c = f_n^{-1}(v_c)$ and $v_c = f_n(d_c)$. And, where v_c is the sum of the cardinals of subset n-tuples, it will be proved that:

$$\forall n, d_c \in \mathbb{N}, \exists v_c, v_{c_i} \in \mathbb{N}: \quad d_c^n = v_c = \sum_{i=1}^m v_{c_i} \quad \Rightarrow \quad d^n = v = \sum_{i=1}^m \left(\prod_{j=1}^n s_{i,j}\right). \quad (3)$$

Every volume is the addition and subtraction of volumes, where each $s_{i,j}$ is $\pm$ -signed. The $n = 2$ case is the dot product, where each $s_{i,j}$ is $\pm$ -signed and $|d|$ is the norm [7]:

$$d^2 = \sum_{i=1}^m \left(\prod_{j=1}^{n=2} s_{i,j}\right) \quad \wedge \quad \exists a_i, b_i \in \mathbb{R}: \quad s_{i,1} = a_i, \quad s_{i,2} = b_i \quad \Rightarrow \quad d^2 = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \quad (4)$$

And, where each summed volume, v_{c_i} , is also the bijective function, $d_{c_i}^n = v_{c_i}$:

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \quad \Rightarrow \quad d^n = \sum_{i=1}^m d_i^n. \quad (5)$$

$|d|$ is the p -norm (Minkowski distance) [11], which will be proved to imply the metric space properties [6, 7]. The $n = 2$ case is the Euclidean distance (flat distance metric) [12,

13]. Multiplying each summed infinitesimal area, ds_i^2 , by a scalar function, $\alpha_i : dr^2 = \sum_{i=1}^m (\alpha_i ds_i^2)$, is a Gaussian-like curved distance that motivated the spacetime curvature metric, $g_{\mu,\nu}$, in Einstein's general relativity theory [3].

Second reduction step

One property salient to physics: All distances spanning a volume are bijective functions of a volume, $v \in \mathbb{R}$, which implies the dimension of each type of volume-spanning distance (for example, the Euclidean distance dimension) is always a $\subseteq \mathbb{R}$.

A second property salient to physics: The commutative property of the sum and multiplication operations defining the number n-tuples and corresponding Euclidean volume (2) allow n number of n-tuple domain sets and volume domain values to be sequenced in all $n!$ permutations, for example, $v = width \times height \times depth = depth \times width \times height = \dots$. That is, the n-tuple domain sets and volume domain values are unordered (no domain element has an intrinsic relative position property).

A third property salient to physics: Starting with any element, all n number of elements can be linearly sequenced via the successor and predecessor functions, $successor(x_i) = x_{i+1}$ and $predecessor(x_i) = x_{i-1}$, for example, $v = \prod_{i=n}^1 s_i$.

Repeatable sequencing of unordered elements requires assigning an immutable linear order (an order that does not change). Where the order of the set elements is immutable, only a cyclic order (list), where $successor(x_n) = x_1$ and $predecessor(x_1) = x_n$, allows starting at each element and linearly sequencing n elements in a successor permutation and a predecessor permutation. For example, where $n = 3$, the volume cyclic successor and predecessor sequences are: $[s_1, s_2, s_3]$ and $[s_1, s_3, s_2]$; $[s_2, s_3, s_1]$ and $[s_2, s_1, s_3]$; $[s_3, s_1, s_2]$ and $[s_3, s_2, s_1]$.

Cyclic lists that allow the linear sequencing of n number of elements in all $n!$ permutations, is where each list element is cyclic adjacent (either an *immediate* cyclic successor or an *immediate* cyclic predecessor) to every other element, referred to, in this article, as an All Adjacent Cyclic List (AACL). For example, the cyclic permutations, $[x_2, x_3, x_1]$ and $[x_2, x_1, x_3]$, require x_1 and x_3 be cyclic adjacent. An AACL will be proved to contain $n \leq 3$ elements.

Third reduction step

Application of each volume spanning distance dimension $\subseteq \mathbb{R}$ property and the AACL property of volume domain elements to physics: There is an AACL of 3 volume domain value dimensions, each $\subseteq \mathbb{R}$, an AACL of 3 non-volume dimensions (time, mass, and charge), each dimension $\subseteq \mathbb{R}$, and one volume-spanning Euclidean distance dimension $\subseteq \mathbb{R}$.

Where the Euclidean distance, time, mass and charge dimensions are divided into the same number of constant-sized intervals, each constant Euclidean distance interval length corresponds to constant interval lengths of time, mass, and charge. 2 consequences are a set of 3 constant direct proportion ratios and a set of 3 constant inverse proportion ratios of Euclidean distance to time, mass, and charge.

The gravity [1, 2], relativity [3, 4], Schwarzschild black hole metric [14, 15], charge equations [2], electromagnetism equations [2], and related constants are derived from the 3 direct proportion ratios. The Planck-Einstein relation [2], Compton wavelength [2], Schrödinger wave equation [5], Dirac wave equation [16], and related constants are derived from combining some of the direct and inverse proportion ratios. And the inverse proportion ratios are used to add quantum extensions to Scharwzchild's metric, Newton's gravity force equation, and Coulomb's charge force equation.

VOLUME

Euclidean volume

Theorem .1. *Euclidean volume,*

$$\forall v_c, d_c, |x_i| \in \{0, \mathbb{N}\}, x_i \in \{x_1, \dots, x_n\}, v_c = \prod_{i=1}^n |x_i| \Rightarrow v = \prod_{i=1}^n s_i, \quad s_i, v \in \mathbb{R}. \quad (6)$$

The formal proof, “Euclidean_volume,” is in the Rocq file, euclidrelations.v.

Proof.

$$\forall v_c, |x_i| \in \{0, \mathbb{N}\}, \kappa \in \mathbb{R} : v_c = \prod_{i=1}^n |x_i| \Leftrightarrow v_c \kappa = (\prod_{i=1}^n |x_i|) \kappa = \prod_{i=1}^n (|x_i| \kappa^{1/n}). \quad (7)$$

$$v_c \kappa = \prod_{i=1}^n (|x_i| \kappa^{1/n}) \wedge \exists v, s_i \in \mathbb{R} : v = v_c \kappa \wedge s_i = |x_i| \kappa^{1/n} \Rightarrow v = \prod_{i=1}^n s_i. \quad \square \quad (8)$$

Lemma .2. *The number of n -tuples, v_c , is the sum of the number of n -tuples, v_{c_i} , in each disjoint subset of n -tuples, implies a volume is the sum of volumes,*

$$v_c = \sum_{i=1}^m v_{c_i} \Rightarrow v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}.$$

The formal proof, “sum_of_volumes,” is in the Rocq file, euclidrelations.v.

Proof. From the condition of this theorem and the Euclidean volume theorem (1):

$$\forall \kappa \in \mathbb{R} : v_c = \sum_{i=1}^m v_{c_i} \Rightarrow v_c \kappa = (\sum_{i=1}^m v_{c_i}) \kappa = \sum_{i=1}^m (v_{c_i} \kappa). \quad (9)$$

$$v_c \kappa = \sum_{i=1}^m (v_{c_i} \kappa) \quad \wedge \quad v = v_c \kappa \quad \wedge \quad v_i = v_{c_i} \kappa \Rightarrow v = \sum_{i=1}^m v_i. \quad \square \quad (10)$$

DISTANCE

Definition .3. Bijective, countable domain value, d_c :

$$\exists v_c, d_c, |x_i| \in \{0, \mathbb{N}\}, \quad x_i \in \{x_1, \dots, x_n\}, \quad v_c = \prod_{i=1}^n |x_i| = \prod_{i=1}^n d_c = d_c^n. \quad (11)$$

Dot product

Theorem .4. *Sum of volumes distance:*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} \Rightarrow d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}).$$

The formal proof, “sum_of_volumes_distance,” is in the Rocq file, euclidrelations.v.

Proof. Multiply both sides of the assumption by κ and apply the Euclidean volume proof (1) to v_i :

$$v_c = \sum_{i=1}^m v_{c_i} \Rightarrow v = \sum_{i=1}^m v_i, \quad v, v_i \in \mathbb{R}. \quad (12)$$

$$v = \sum_{i=1}^m v_i \quad \wedge \quad v_i = \prod_j^n s_{i,j} \Rightarrow v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \quad (13)$$

$$d_c^n = v_c \Rightarrow d_c^n \kappa = (d_c \kappa^{1/n})^n = v_c \kappa. \quad (14)$$

$$\exists d, v \in \mathbb{R} : d = d_c \kappa^{1/n} \quad \wedge \quad v = v_c \kappa \quad \wedge \quad (d_c \kappa^{1/n})^n = v_c \kappa \Rightarrow d^n = v. \quad (15)$$

$$d^n = v \quad \wedge \quad v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}) \Rightarrow d^n = v = \sum_{i=1}^m (\prod_{j=1}^n s_{i,j}). \quad \square \quad (16)$$

The $n = 2$ case is the dot product, where each $s_{i,j}$ is $\pm$ -signed, and $|d|$ is the norm [7]:

$$d^2 = \sum_{i=1}^m (\prod_{j=1}^{n=2} s_{i,j}) \wedge \exists a_{i,1}, a_{i,2} \in \mathbb{R} : s_{i,1} = a_i, s_{i,2} = b_i \Rightarrow d^2 = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \quad (17)$$

Minkowski distance (p -norm)

Theorem .5. *Minkowski distance (p -norm):*

$$d_c^n = v_c = \sum_{i=1}^m v_{c_i} = \sum_{i=1}^m d_{c_i}^n \Leftrightarrow d^n = \sum_{i=1}^m d_i^n.$$

The formal proof, “Minkowski_distance,” is in the Rocq file, euclidrelations.v.

Proof. From the sum of volumes distance theorem .4:

$$d_{c_i}^n = v_{c_i} \Rightarrow d_i^n = v_i. \quad (18)$$

$$d^n = v = \sum_{i=1}^m v_i \quad \wedge \quad d_i^n = v_i \Rightarrow d^n = \sum_{i=1}^m d_i^n \quad \square \quad (19)$$

Distance inequalities

Theorem .6. *Distance triangle inequality*

$$\forall n \in \mathbb{N}, v_a, v_b \geq 0 : (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}.$$

The formal proof, distance_inequality, is in the Rocq file, euclidrelations.v.

Proof. Expand $(v_a^{1/n} + v_b^{1/n})^n$ using the binomial expansion:

$$\begin{aligned} \forall v_a, v_b \geq 0 : \quad v_a + v_b &\leq v_a + v_b + \sum_{i=1}^n \binom{n}{i} (v_a^{1/n})^{n-i} (v_b^{1/n})^i \\ &\quad + \sum_{i=1}^n \binom{n}{i} (v_a^{1/n})^i (v_b^{1/n})^{n-i} = (v_a^{1/n} + v_b^{1/n})^n. \end{aligned} \quad (20)$$

Take the n^{th} root of both sides of the inequality 20:

$$\forall v_a, v_b \geq 0, n \in \mathbb{N} : \quad v_a + v_b \leq (v_a^{1/n} + v_b^{1/n})^n \Rightarrow (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n}. \quad (21)$$

$\square$

Theorem .7. *Distance sum inequality*

$$\forall m, n \in \mathbb{N}, a_i, b_i \geq 0 : (\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}.$$

The formal proof, distance_sum_inequality, is in the Rocq file, euclidrelations.v.

Proof. Apply the distance triangle inequality (.6):

$$\begin{aligned} \forall m, n \in \mathbb{N}, v_a, v_b \geq 0 : \quad v_a &= \sum_{i=1}^m a_i^n \quad \wedge \quad v_b = \sum_{i=1}^m b_i^n \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n} \\ \Rightarrow \quad ((\sum_{i=1}^m a_i^n) &+ (\sum_{i=1}^m b_i^n))^{1/n} = (\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}. \quad \square \end{aligned} \quad (22)$$

Metric Space

All Minkowski distances (p -norms) imply the metric space properties. The formal proofs: triangle inequality, symmetry, non-negativity, and identity of indiscernibles are in the Rocq file, euclidrelations.v.

Theorem .8. *Triangle Inequality:*

$$\begin{aligned} d(s_1, s_2) &= (\sum_{i=1}^2 s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k \\ \Rightarrow \quad d(u, w) &= (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \end{aligned}$$

Proof. $\forall p \geq 1, \quad u = s_1, \quad w = s_2, \quad v = w/k$:

$$(u^p + w^p)^{1/p} \leq ((u^p + w^p) + 2v^p)^{1/p} = ((u^p + v^p) + (v^p + w^p))^{1/p}. \quad (23)$$

Apply the distance triangle inequality (6) to the inequality 23:

$$\begin{aligned} (u^p + w^p)^{1/p} &\leq ((u^p + v^p) + (v^p + w^p))^{1/p} \quad \wedge \quad (v_a + v_b)^{1/n} \leq v_a^{1/n} + v_b^{1/n} \\ &\quad \wedge \quad v_a = u^p + v^p \quad \wedge \quad v_b = v^p + w^p \\ \Rightarrow \quad (u^p + w^p)^{1/p} &\leq ((u^p + v^p) + (v^p + w^p))^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} \\ \Rightarrow \quad d(u, w) &= (u^p + w^p)^{1/p} \leq (u^p + v^p)^{1/p} + (v^p + w^p)^{1/p} = d(u, v) + d(v, w). \quad \square \quad (24) \end{aligned}$$

Theorem .9. *Symmetry:*

$$d(s_1, s_2) = (\sum_{i=1}^2 s_i^p)^{1/p}, \quad p \geq 1, \quad u = s_1, \quad v = s_2 \quad \Rightarrow \quad d(u, v) = d(v, u).$$

Proof. By the commutative law of addition:

$$\begin{aligned} \forall p : p \geq 1, \quad d(s_1, s_2) &= (\sum_{i=1}^2 s_i^p)^{1/p} = (s_1^p + s_2^p)^{1/p} \\ \Rightarrow \quad d(u, v) &= (u^p + v^p)^{1/p} = (v^p + u^p)^{1/p} = d(v, u). \quad \square \quad (25) \end{aligned}$$

Theorem .10. *Non-negativity:*

$$d(s_1, s_2) = (\sum_{i=1}^2 s_i^p)^{1/p}, \quad p \in \mathbb{N} \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(u, v) \geq 0.$$

Proof.

$$\forall n \geq 1 \quad \wedge \quad s_1, s_2 \geq 0 \quad \Rightarrow \quad d(s_1, s_2) = (s_1^p + s_2^p)^{1/p} \geq 0. \quad (26)$$

$$\forall u = s_1, v = s_2 \quad \wedge \quad d(s_1, s_2) = (s_1^p + s_2^p)^{1/p} \geq 0 \quad \Rightarrow \quad d(u, v) = (u^p + v^p)^{1/p} \geq 0. \quad \square \quad (27)$$

Theorem .11. *Identity of Indiscernibles:* $d(u, u) = 0$.

Proof. From the non-negativity property (.10):

$$d(u, w) \geq 0 \quad \wedge \quad d(u, v) \geq 0 \quad \wedge \quad d(v, w) \geq 0 \quad (28)$$

$$\Rightarrow \quad \exists d(u, w) = d(u, v) = d(v, w) = 0. \quad (29)$$

$$d(u, w) = d(v, w) = 0 \quad \Rightarrow \quad u = v. \quad (30)$$

$$d(u, v) = 0 \quad \wedge \quad u = v \quad \Rightarrow \quad d(u, u) = 0. \quad \square \quad (31)$$

PROPERTIES LIMITING A SET TO AT MOST 3 ELEMENTS

The following definitions and proof use first order logic. A Horn clause-like expression is used, here, to make the proof easier to read. By convention, the proof goal, in a Horn clause, is on the left side of the implication sign ($\leftarrow$) and supporting facts are on the right side. The formal proofs in the Rocq file, `threed.v`, are: `adj111`, `adj122`, `adj212`, `adj123`, `adj133`, `adj233`, `adj213`, `adj313`, `adj323`, and `not_all_mutually_adjacent_gt_3`.

Definition .12. Immediate Cyclic Successor of m is n :

$$\forall x_m, x_n \in \{x_1, \dots, x_{\text{setsize}}\} :$$

$$\text{Successor}(m, n, \text{setsize}) \leftarrow (m = \text{setsize} \wedge n = 1) \vee (n = m + 1 \wedge n \leq \text{setsize}). \quad (32)$$

Definition .13. Immediate Cyclic Predecessor of m is n :

$$\forall x_m, x_n \in \{x_1, \dots, x_{\text{setsize}}\} :$$

$$\text{Predecessor}(m, n, \text{setsize}) \leftarrow (m = 1 \wedge n = \text{setsize}) \vee (n = m - 1 \wedge n \geq 1). \quad (33)$$

Definition .14. Adjacent: Element m is sequentially cyclic adjacent to element n if the immediate cyclic successor of m is n or the immediate cyclic predecessor of m is n :

$$\forall x_m, x_n \in \{x_1, \dots, x_{\text{setsize}}\} :$$

$$\text{Adjacent}(m, n, \text{setsize}) \leftarrow \text{Successor}(m, n, \text{setsize}) \vee \text{Predecessor}(m, n, \text{setsize}). \quad (34)$$

Definition .15. All Adjacent Cyclic List (AACL), where every set element is cyclic adjacent to every other element):

$$\forall x_m, x_n \in \{x_1, \dots, x_{\text{setsize}}\} : \quad \text{Adjacent}(m, n, \text{setsize}). \quad (35)$$

Theorem .16. *An AACL is limited to at most 3 elements.*

$$\forall x_m, x_n \in \{x_1, \dots, x_{setsize}\} : \quad Adjacent(m, n, setsize) \Rightarrow setsize \leq 3. \quad (36)$$

Proof.

Every element is adjacent to every other element, where $setsize \in \{1, 2, 3\}$:

$$Adjacent(1, 1, 1) \leftarrow Successor(1, 1, 1) \leftarrow (m = setsize \wedge n = 1). \quad (37)$$

$$Adjacent(1, 2, 2) \leftarrow Successor(1, 2, 2) \leftarrow (n = m + 1 \wedge n \leq setsize). \quad (38)$$

$$Adjacent(1, 2, 3) \leftarrow Successor(1, 2, 3) \leftarrow (n = m + 1 \wedge n \leq setsize). \quad (39)$$

$$Adjacent(2, 1, 3) \leftarrow Predecessor(2, 1, 3) \leftarrow (n = m - 1 \wedge n \geq 1). \quad (40)$$

$$Adjacent(3, 1, 3) \leftarrow Successor(3, 1, 3) \leftarrow (n = setsize \wedge m = 1). \quad (41)$$

$$Adjacent(1, 3, 3) \leftarrow Predecessor(1, 3, 3) \leftarrow (m = 1 \wedge n = setsize). \quad (42)$$

$$Adjacent(2, 3, 3) \leftarrow Successor(2, 3, 3) \leftarrow (n = m + 1 \wedge n \leq setsize). \quad (43)$$

$$Adjacent(3, 2, 3) \leftarrow Predecessor(3, 2, 3) \leftarrow (n = m - 1 \wedge n \geq 1). \quad (44)$$

Element 2 is the only immediate successor of element 1 for all $setsize \geq 3$, which implies element 3 is not ($\neg$) an immediate successor of element 1 for all $setsize \geq 3$:

$$\begin{aligned} &\neg Successor(1, 3, setsize \geq 3) \\ &\quad \leftarrow Successor(1, 2, setsize \geq 3) \leftarrow (n = m + 1 \leq setsize). \end{aligned} \quad (45)$$

Element $n = setsize > 3$ is the only immediate predecessor of element 1, which implies element 3 is not ($\neg$) an immediate predecessor of element 1 for all $setsize > 3$:

$$\begin{aligned} &\neg Predecessor(1, 3, setsize \geq 3) \\ &\quad \leftarrow Predecessor(1, setsize, setsize > 3) \leftarrow (m = 1 \wedge n = setsize > 3). \end{aligned} \quad (46)$$

For all $setsize > 3$, some elements are not ($\neg$) sequentially adjacent to every other element (not immediate symmetric):

$$\begin{aligned} &\neg Adjacent(1, 3, setsize > 3) \\ &\quad \leftarrow \neg Successor(1, 3, setsize > 3) \quad \wedge \quad \neg Predecessor(1, 3, setsize > 3). \quad \square \end{aligned} \quad (47)$$

The Symmetric goal matches the Adjacent goals 37 and fails for all “setsize” greater than three.

DERIVATION OF PHYSICS EQUATIONS AND CONSTANTS

3 direct and 3 inverse proportion ratios

Application of the AACL property of volume (.16) and the Euclidean distance dimension $\subseteq \mathbb{R}$ to physics (.5): $\{r_1, r_2, r_3\}$ is an AACL of 3 volume dimensions, each dimension having the $\subseteq \mathbb{R}$ set membership property and $\{t \text{ (time)}, m \text{ (mass)}, q \text{ (charge)}\}$ is an AACL of 3 “non-volume” dimensions, each dimension $\subseteq \mathbb{R}$. There is a total of 6 AACL dimensions with the $\subseteq \mathbb{R}$ property: $\{r_1, r_2, r_3\}$ and $\{t, m, q\}$.

Where the Euclidean distance, time, mass, and charge dimensions are each divided into n number of constant-sized interval lengths, for i in 1 to n , the i^{th} constant unit interval length, r_p , of Euclidean distance corresponds to the i^{th} constant unit interval lengths: t_p of time, m_p of mass, and q_p of charge, where “unit” means a measurement type, for example, meters, seconds, kilograms, and Coulombs.

Two consequences of the Euclidean distance and AACL, dimensions $\subseteq \mathbb{R}$, and corresponding constant interval lengths are:

1. A set of 3 constant direct proportion ratios and proportion constants:

$$\forall r_p, t_p, m_p, q_p, k_1, k_2, k_3 \in \mathbb{R}, \exists r, t, m, q \in \mathbb{R} : \quad \frac{r_p}{t_p} = \frac{r_p k_1}{t_p k_1} = \frac{r}{t} \quad \wedge \quad \frac{r_p}{m_p} = \frac{r_p k_2}{m_p k_2} = \frac{r}{m} \quad \wedge \quad \frac{r_p}{q_p} = \frac{r_p k_3}{q_p k_3} = \frac{r}{q}. \quad (48)$$

$$\begin{aligned} \exists c, c_t, c_m, c_q \in \mathbb{R} : \quad r_p/t_p = c_t = c \quad \wedge \quad r_p/m_p = c_m \quad \wedge \quad r_p/q_p = c_q \\ \Rightarrow \quad r/t = r_p/t_p = c_t = c \quad \wedge \quad r/m = r_p/m_p = c_m \quad \wedge \quad r/q = r_p/q_p = c_q. \end{aligned} \quad (49)$$

2. A set of 3 constant inverse proportion ratios and proportion constants:

$$\begin{aligned} \forall r_p, t_p, m_p, q_p, k_1, k_2, k_3 \in \mathbb{R}, \exists r, t, m, q \in \mathbb{R} : \\ r_p t_p = r_p t_p \left(\frac{k_1}{k_1} \right) = \left(\frac{r_p}{k_1} \right) (t_p k_1) = r t \quad \wedge \quad r_p m_p = r_p m_p \left(\frac{k_2}{k_2} \right) = \left(\frac{r_p}{k_2} \right) (m_p k_2) = r m \\ \wedge \quad r_p q_p = r_p q_p \left(\frac{k_3}{k_3} \right) = \left(\frac{r_p}{k_3} \right) (q_p k_3) = r q. \end{aligned} \quad (50)$$

$$\begin{aligned} \exists k_t, k_m, k_q \in \mathbb{R} : \quad r_p t_p = k_t \quad \wedge \quad r_p m_p = k_m \quad \wedge \quad r_p q_p = k_q \\ \Rightarrow \quad r t = r_p t_p = k_t \quad \wedge \quad r m = r_p m_p = k_m \quad \wedge \quad r q = r_p q_p = k_q. \end{aligned} \quad (51)$$

G, Newton, Gauss, Poisson gravity laws

From equations 49, linear acceleration, r/t^2 , can be related to mass, m , and distance, r :

$$r = c_m m \quad \wedge \quad r = c_t t \quad \Rightarrow \quad \frac{r}{(c_t t)^2} = \frac{c_m m}{r^2} \quad \Rightarrow \quad \frac{r}{t^2} = \frac{(c_m c_t^2) m}{r^2} = \frac{Gm}{r^2}. \quad (52)$$

Where r_p , t_p , and m_p are the Planck length, time, and mass units, $G = c_m c_t^2 = r_p^3/m_p t_p^2 \approx 6.67430 \cdot 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2}$, which are the empirical value and SI units of G [17].

Newton's gravity law [1] follows from multiplying both sides of equation 52 by m :

$$\frac{r}{t^2} = \frac{Gm}{r^2} \quad \Leftrightarrow \quad F \stackrel{\text{def}}{=} \frac{mr}{t^2} = \frac{Gm^2}{r^2}. \quad (53)$$

$$F = \frac{Gm^2}{r^2} \quad \wedge \quad \forall m \in \mathbb{R} : \exists m_1, m_2 \in \mathbb{R} : m_1 m_2 = m^2 \quad \Rightarrow \quad F = \frac{Gm_1 m_2}{r^2}. \quad (54)$$

A new interpretation of the Gauss and Poisson gravity laws is presented here. Equation 52 relates linear acceleration, r/t^2 , to mass and distance. Gauss's gravity vector field, $\mathbf{g}$, and Poisson's gravity scalar field, $-\nabla\Phi$, is the angular acceleration, $2\pi r/t^2$. Applying equation 52, where the minus sign is the vector direction (attractive force):

$$\frac{|\mathbf{r}|}{t^2} = -\frac{Gm}{|\mathbf{r}|^2} \quad \wedge \quad \mathbf{g} = -\nabla\Phi = \frac{2\pi|\mathbf{r}|}{t^2} \quad \Rightarrow \quad \mathbf{g} = -\nabla\Phi = -\frac{2\pi Gm}{|\mathbf{r}|^2}. \quad (55)$$

$$\mathbf{g} = -\nabla\Phi = -\frac{2\pi Gm}{|\mathbf{r}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{g} = \nabla^2\Phi = \frac{4\pi Gm}{|\mathbf{r}|^3} = 4\pi G\rho, \quad (56)$$

which are the differential forms of the Gauss and Poisson gravity laws [2], where $\rho = m/|\mathbf{r}|^3$ is the mass density.

Special relativity

$$\begin{aligned} \forall \tau \in \{t, m, q\}, \quad r^2 &= r'^2 + s^2, \quad \exists \mu_\tau, \nu_\tau \in \mathbb{R} : r = \mu_\tau \tau \quad \wedge \quad s = \nu_\tau \tau \\ \Rightarrow \quad (\mu_\tau \tau)^2 &= r'^2 + (\nu_\tau \tau)^2 \quad \Rightarrow \quad r' = \sqrt{(\mu_\tau \tau)^2 - (\nu_\tau \tau)^2} = \mu_\tau \tau \sqrt{1 - (\nu_\tau / \mu_\tau)^2}. \end{aligned} \quad (57)$$

Local frame distance, r' , contracts relative to a distant frame distance, r , as $\nu_\tau \rightarrow \mu_\tau$:

$$r' = \mu_\tau \tau \sqrt{1 - (\nu_\tau / \mu_\tau)^2} \quad \wedge \quad \mu_\tau \tau = r \quad \Rightarrow \quad r' = r \sqrt{1 - (\nu_\tau / \mu_\tau)^2}. \quad (58)$$

Where τ' is time in the local frame, less time lapses (time dilates) relative to time, τ , in the distant (coordinate) frame: τ , as $\nu_\tau \rightarrow \mu_\tau$:

$$\mu_\tau \tau = r' / \sqrt{1 - (\nu_\tau / \mu_\tau)^2} \quad \wedge \quad r' = \mu_\tau \tau \quad \Rightarrow \quad \tau' = \tau \sqrt{1 - (\nu_\tau / \mu_\tau)^2}. \quad (59)$$

Where τ is type, time, the space-like flat Minkowski spacetime event interval is:

$$\begin{aligned} dr^2 = dr'^2 + ds^2 \quad \wedge \quad ds^2 = dr_1^2 + dr_2^2 + dr_3^2 \quad \wedge \quad d(\mu_\tau \tau) = dr \\ \Rightarrow \quad dr'^2 = d(\mu_\tau \tau)^2 - dr_1^2 - dr_2^2 - dr_3^2. \end{aligned} \quad (60)$$

Einstein's general relativity

The Einstein field equation (EFE) is:

$$\mathbf{G}_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}, \quad \mathbf{G}_{\mu,\nu} = \mathbf{R} - \frac{1}{2} R g_{\mu,\nu}. \quad (61)$$

where: $\mathbf{G}_{\mu,\nu}$ is Einstein's tensor, $\mathbf{R}$ is the Ricci curvature, R is the scalar curvature, $g_{\mu,\nu}$ is the metric tensor, Λ is a data-fitting constant, $\kappa = 8\pi G/c^4$, is Einstein's constant, and $T_{\mu,\nu}$ is the stress-energy-momentum tensor [3, 4].

The EFE extends the, previously derived, Poisson's law of gravity, $\nabla^2 \Phi = 4\pi G m / |\mathbf{r}|^3$ (56), to include a time component. In 3-dimensional space, each component of $\mathbf{G}_{m,n}$ ($m, n, i \in \{1, 2, 3\}$) is the Gaussian curvature, where, for example, $\mathbf{G}_{i,i} = \nabla^2 \Phi = 4\pi G m / |r_i|^3$.

In 4-dimensional spacetime, the Ricci curvature of each component of $\mathbf{G}_{\mu,\nu}$ ($\mu, \nu \in \{0, 1, 2, 3\}$) has a time curvature, $\nabla^2 \Phi(ct)$, plus a space curvature, $\nabla^2 \Phi(\mathbf{r})$. And the Minkowski curved spacetime interval component for time is $\mathbf{G}_{0,0}(cdt)^2 = c^2 \mathbf{G}_{0,0} dt^2$:

$$\begin{aligned} |\mathbf{r}| = r = ct \quad \wedge \quad c^2 \mathbf{G}_{0,0} = \nabla^2 \Phi(ct) + \nabla^2 \Phi(\mathbf{r}) \quad \Rightarrow \quad c^2 \mathbf{G}_{0,0} = 2 \nabla^2 \Phi(\mathbf{r}) \quad (62) \\ \Rightarrow \quad \mathbf{G}_{0,0} = \frac{2}{c^2} \nabla^2 \Phi(\mathbf{r}) = \frac{2}{c^2} \cdot \frac{4\pi G m}{|\mathbf{r}|^3} = \frac{8\pi G}{c^2} \cdot \frac{m}{|\mathbf{r}|^3} \cdot \frac{c^2}{c^2} = \frac{8\pi G}{c^4} \cdot \frac{mc^2}{|\mathbf{r}|^3} = \kappa \mathbf{T}_{0,0}, \end{aligned} \quad (63)$$

where $\kappa = 8\pi G/c^4$ is Einstein's constant and $\mathbf{T}_{0,0}$ is the (kinetic plus potential) energy density component of $\mathbf{T}_{\mu,\nu}$ [4]. Using Einstein's constant, κ , for all components of $\mathbf{G}_{\mu,\nu}$ and adding in the fudge factor, $\Lambda g_{\mu,\nu}$, yields the EFE: $\mathbf{G}_{\mu,\nu} + \Lambda g_{\mu,\nu} = \kappa T_{\mu,\nu}$.

Analytic general relativity solutions

The common perspective is that the EFE is the law of gravity, where the non-linearity makes finding exact (analytical) solutions difficult [4]. Here, a different perspective is taken, where: 1) The laws of physics determine the EFE: determines both the spacetime curvature, $\mathbf{G}_{\mu,\nu}$, and the distribution of stress-energy-momentum, $\mathbf{T}_{\mu,\nu}$. 2) The laws of physics

are specified as the components of the metric, $g_{\mu,\nu}$. 3) $g_{\mu,\nu}$ can be determined exactly, independent of the EFE, for those cases where the laws (equations) can be derived from the ratios. Derivation of the Schwarzschild metric from the ratios, in the next subsection, is an example.

Step 1) Use the unit ratios and previously derived equations (laws) , for example, the special relativity equations (58) and the ratio-derived constant, G (52), to derive the scalar functions (laws) assigned to the components of $g_{\mu,\nu}$.

Step 2) The 4D flat spacetime interval equation is an instance of the 2D equation (60). Therefore, derive the metric first as a 2D metric and expand later to a 4D metric.

(Optional step 2d) The 2D metric tensor uses the much simpler 2D Ricci curvature (which, in 2D, is the Gaussian curvature). And the 2D tensors reduce the number of independent equations to solve. The 2D solutions can next be used to set constraints on the solutions in the 4D tensor, much like an Euclidean distance value places constraints on the coordinate distance values.

Step 3) Translate from 2D to 4D. For spherically symmetric cases, the simplest method to translate from 2D to 4D is to use spherical coordinates, where the first 2 diagonal components of $g_{\nu,\mu}$ remain unchanged and the other 2 diagonal components are functions of the longitude angle, θ , and latitude angle, ϕ .

Schwarzschild's time dilation and metric

From equations 58 and 49:

$$r = c_m m \Rightarrow c_m m / r = 1 \Rightarrow \sqrt{1 - (v^2/c^2)} = \sqrt{1 - (1)(v^2/c^2)} = \sqrt{1 - c_m m v^2 / r c^2}. \quad (64)$$

Where v_{escape} is the escape velocity and KE is the kinetic energy:

$$\begin{aligned} \sqrt{1 - (v^2/c^2)} &= \sqrt{1 - c_m m v^2 / r c^2} \quad \wedge \quad KE = m v^2 / 2 = m v_{escape}^2 / 2 \\ &\Rightarrow \sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2 c_m m v_{escape}^2 / r c^2}. \end{aligned} \quad (65)$$

For a photon, the escape velocity, $v_{escape} = c$.

$$\begin{aligned} \sqrt{1 - (v^2/c^2)} &= \sqrt{1 - 2 c_m m c^2 / r c^2} \quad \wedge \quad v_{escape} = c \\ &\Rightarrow \sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2 c_m m c^2 / r c^2}. \end{aligned} \quad (66)$$

Combining equation 66 with the derivation of G (52):

$$c_m c^2 = G \wedge \sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2c_m m c^2 / r c^2} \Rightarrow \sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2Gm/rc^2}. \quad (67)$$

Schwarzschild defined the black hole event horizon radius as $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$ [14, 15]. Here, α , is called the Schwarzschild radius, $r_s \stackrel{\text{def}}{=} 2Gm/c^2$:

$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - 2Gm/rc^2} \wedge r_s \stackrel{\text{def}}{=} 2Gm/c^2 \Rightarrow \sqrt{1 - (v^2/c^2)} = \sqrt{1 - r_s/r}. \quad (68)$$

Combining equation 68 with equation 59 yields Schwarzschild's gravitational time dilation:

$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - r_s/r} \wedge t' = t\sqrt{1 - (v^2/c^2)} \Rightarrow t' = t\sqrt{1 - r_s/r}. \quad (69)$$

Combining equations 58 and 68.

$$r' = r\sqrt{1 - (v/c)^2} \wedge \sqrt{1 - (v^2/c^2)} = \sqrt{1 - r_s/r} \Rightarrow r' = r\sqrt{1 - r_s/r}. \quad (70)$$

Applying equation 70 to the time-like spacetime interval equation 60, where $ds^2 < 0$:

$$\begin{aligned} r' &= r\sqrt{1 - r_s/r} \wedge ds^2 = dr'^2 - dr^2 \\ \Rightarrow ds^2 &= (\sqrt{1 - r_s/r} dr)^2 - (dr'/\sqrt{1 - r_s/r})^2 = (1 - r_s/r)dr^2 - (1 - r_s/r)^{-1}dr'^2. \end{aligned} \quad (71)$$

$$\begin{aligned} ds^2 &= (1 - r_s/r)dr^2 - (1 - r_s/r)^{-1}dr'^2 \wedge dr = d(ct) \wedge c = 1 \\ \Rightarrow ds^2 &= (1 - r_s/r)dt^2 - (1 - r_s/r)^{-1}dr'^2. \end{aligned} \quad (72)$$

Using spherical coordinates to translate from 2D to 4D yields the $+ - - -$ form of Schwarzschild's black hole metric, $g_{\mu,\nu}$ [14, 15]:

$$\begin{aligned} ds^2 &= (1 - r_s/r)dt^2 - (1 - r_s/r)^{-1}dr'^2 \quad (\text{in 2D}) \\ \Rightarrow ds^2 &= (1 - r_s/r)dt^2 - (1 - r_s/r)^{-1}dr'^2 - r^2 d\theta^2 - r^2 \sin^2 \theta d\phi^2 \quad (\text{in 4D}) \\ \Rightarrow g_{\mu,\nu} &= \text{diag}[(1 - r_s/r), -(1 - r_s/r)^{-1}, -r^2, -r^2 \sin^2 \theta]. \end{aligned} \quad (73)$$

$k_e, \varepsilon_0, \mathbf{E}$, Coulomb and Gauss laws

From equations 49, linear acceleration, r/t^2 , can be related to charge, q , and distance, r :

$$r = c_q q \wedge r = c_t t \Rightarrow \frac{r}{(c_t t)^2} = \frac{c_q q}{r^2} \Rightarrow \frac{r}{t^2} = \frac{(c_q c_t^2) q}{r^2}. \quad (74)$$

$$r = c_m m \wedge r = c_q q \Rightarrow c_m m = c_q q \Rightarrow m = (c_q/c_m) q. \quad (75)$$

Combining equations 74 and 75 yields Coulombs's law and constant, k_e , [2]:

$$\frac{r}{t^2} = \frac{(c_q c_t^2)q}{r^2} \quad \wedge \quad m = (c_q/c_m)q \quad \Rightarrow \quad F \stackrel{\text{def}}{=} \frac{mr}{t^2} = \frac{(c_q^2 c_t^2/c_m)q^2}{r^2} = \frac{k_e q^2}{r^2}. \quad (76)$$

Where r_p , t_p , m_p , and q_p are the Planck units, $k_e = c_q^2 c_t^2/c_m = m_p r_p^3/t_p^2 q_p^2 \approx 8.987552 \cdot 10^9 \text{ kg} \cdot \text{m}^3 \cdot \text{s}^{-2} \cdot \text{C}^{-2} \stackrel{\text{def}}{=} N \cdot \text{m}^2 \cdot \text{C}^{-2}$, which are the empirical value and SI units [2, 17].

$$\forall q \in \mathbb{R} \exists q_1, q_2 \in \mathbb{R} : |q_1 q_2| = q^2 \quad \wedge \quad F = \frac{k_e q^2}{r^2} \quad \Rightarrow \quad F = \frac{k_e |q_1 q_2|}{r^2}. \quad (77)$$

A new interpretation of Gauss's electric field law is presented here. Equation 74 relates the linear acceleration, r/t^2 , to charge and distance. Gauss's electric field, $\mathbf{E}$, relates angular acceleration, $2\pi r/t^2$ to charge and distance. Applying equation 76:

$$F_C = \frac{mr}{t^2} = \frac{k_e q^2}{r^2} \quad \Rightarrow \quad \exists F_E \in \mathbb{R} : F_E = 2\pi F_C = \frac{2\pi mr}{t^2} = \frac{2\pi k_e q^2}{r^2}. \quad (78)$$

$$F_E = \frac{2\pi k_e q^2}{r^2} = q \left(\frac{2\pi k_e q}{r^2} \right) \quad \wedge \quad \exists E \in \mathbb{R} : E = \frac{2\pi k_e q}{r^2} \quad \Rightarrow \quad F_E = qE. \quad (79)$$

The electric field, $E = 2\pi k_e q/r^2$, conforms to the SI units $\text{kg} \cdot \text{m} \cdot \text{s}^{-2} \cdot \text{C}^{-1} \stackrel{\text{def}}{=} \text{kg} \cdot \text{m} \cdot \text{s}^{-3} \cdot \text{A}^{-1}$ [17].

$$\mathbf{E} = \frac{2\pi k_e q}{|\mathbf{r}|^2} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\mathbf{r}|^3}. \quad (80)$$

$$\nabla \cdot \mathbf{E} = -\frac{4\pi k_e q}{|\mathbf{r}|^3} \quad \wedge \quad \varepsilon_0 \stackrel{\text{def}}{=} \frac{1}{4\pi k_e} \quad \wedge \quad \rho \stackrel{\text{def}}{=} \frac{q}{|\mathbf{r}|^3} \quad \Rightarrow \quad \nabla \cdot \mathbf{E} = -\rho/\varepsilon_0, \quad (81)$$

which is Gauss's electric field law in differential form [2].

B, μ_0 , and Lorentz's law

Applying the distance contraction equation, 58, to equation 79, where r' and r are swapped such that r' is the distant observer frame and r is local (rest) frame of reference:

$$r' = r/\sqrt{1 - v^2/c^2} \quad \wedge \quad F' = \frac{2\pi k_e q^2}{r'^2} \quad \Rightarrow \quad F = \frac{2\pi k_e q^2 (1 - v^2/c^2)}{r^2}. \quad (82)$$

From equation 79:

$$E = \frac{2\pi k_e q}{r^2} \quad \wedge \quad F = \frac{2\pi k_e q^2 (1 - v^2/c^2)}{r^2} \quad \Rightarrow \quad F = q(E - \frac{(2\pi k_e/c^2)q}{r^2} v^2). \quad (83)$$

$$F = q(E - \frac{(2\pi k_e/c^2)q}{r^2} v^2) \quad \wedge \quad \exists B \in \mathbb{R} : B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \Rightarrow \quad F = q(E - Bv), \quad (84)$$

where the magnetic field, $B = (2\pi k_e/c)vq/r^2$, conforms to the base SI units: $kg \cdot C^{-1} \cdot s^{-1} \stackrel{\text{def}}{=} kg \cdot A^{-1} \cdot s^{-2} \stackrel{\text{def}}{=} T$ [17]. And assigning the constant portion to μ_0 :

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \Rightarrow \frac{\partial B}{\partial r} = \frac{(4\pi k_e/c^2)vq}{r^3} = \frac{\mu_0 vq}{r^3}. \quad (85)$$

where $\mu_0 = 4\pi k_e/c^2 = m_p r_p/q_p^2$ conforms to the SI units, $kg \cdot m \cdot C^{-2} \stackrel{\text{def}}{=} kg \cdot m \cdot s^{-2} A^{-2} \stackrel{\text{def}}{=} N \cdot A^{-2}$ [17]. And translating equation 84 to vector form:

$$F = q(E - Bv) \Rightarrow \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \mathbf{v}). \quad (86)$$

$$\mathbf{B} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{B}) \quad \wedge \quad \mathbf{F} = q(\mathbf{E} - \mathbf{B} \times \mathbf{v}) \Rightarrow \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}), \quad (87)$$

which is Lorentz law in the local frame of reference [2].

Faraday's law

From the magnetic field equation 87, where the electric and magnetic fields are propagating at the speed, $v = c$:

$$B = \frac{(2\pi k_e/c^2)vq}{r^2} \quad \wedge \quad v = c \quad \wedge \quad r = ct \Rightarrow B = \frac{(2\pi k_e/c^3)q}{t^2}. \quad (88)$$

$$B = \frac{(2\pi k_e/c^3)q}{t^2} \Rightarrow \frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3}. \quad (89)$$

$$\frac{\partial B}{\partial t} = -\frac{(4\pi k_e/c^3)q}{t^3} \quad \wedge \quad r = ct \Rightarrow \frac{\partial B}{\partial t} = -\frac{4\pi k_e q}{r^3}. \quad (90)$$

Faraday's law is the case, where the electric field, $\mathbf{E}$, is a single radial vector, $\mathbf{E}_{\mathbf{r}_1}$ acting on some point, P . Therefore, at point, P , $\mathbf{E}_{\mathbf{r}_2} = 0$ and $\mathbf{E}_{\mathbf{r}_3} = 0$. From the Lorentz equation (87), the magnetic field, $\mathbf{B}$ at point P , is perpendicular to $\mathbf{E}_{\mathbf{r}_1}$ (for example, aligned on $\mathbf{r}_2$). For a constant force, the Lorentz equation makes the electric field, $\mathbf{E}_{\mathbf{r}_1}$, a function of $\mathbf{B}_{\mathbf{r}_2}$. Combining with equation 80:

$$\begin{aligned} \mathbf{E}_{\mathbf{r}_1} &= f(\mathbf{B}_{\mathbf{r}_2}) = 2\pi k_e q / |\mathbf{r}_2|^2 \quad \wedge \quad \mathbf{E}_{\mathbf{r}_2} = 0 \quad \wedge \quad \mathbf{E}_{\mathbf{r}_3} = 0 \\ \Rightarrow \quad \nabla \times \mathbf{E} &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\partial}{\partial \mathbf{r}_1^2} & \frac{\partial}{\partial \mathbf{r}_2^2} & \frac{\partial}{\partial \mathbf{r}_3^2} \\ \mathbf{E}_{\mathbf{r}_1} & 0 & 0 \end{vmatrix} = \left(\frac{\partial \mathbf{E}_{\mathbf{r}_3}}{\partial \mathbf{r}_2^2} - \frac{\partial \mathbf{E}_{\mathbf{r}_2}}{\partial \mathbf{r}_3^2} \right) \hat{\mathbf{i}} + \left(\frac{\partial \mathbf{E}_{\mathbf{r}_1}}{\partial \mathbf{r}_3^2} - \frac{\partial \mathbf{E}_{\mathbf{r}_3}}{\partial \mathbf{r}_1^2} \right) \hat{\mathbf{j}} + \left(\frac{\partial \mathbf{E}_{\mathbf{r}_2}}{\partial \mathbf{r}_1^2} - \frac{\partial \mathbf{E}_{\mathbf{r}_1}}{\partial \mathbf{r}_2^2} \right) \hat{\mathbf{k}} \\ &= 0\hat{\mathbf{i}} + 0\hat{\mathbf{j}} + \left(0 - \frac{\partial 2\pi k_e q / |\mathbf{r}_2|^2}{\partial \mathbf{r}_2^2} \right) \hat{\mathbf{k}} = \frac{4\pi k_e q}{|\mathbf{r}_2|^3} \hat{\mathbf{k}} = \frac{4\pi k_e q}{|\mathbf{r}|^3}. \end{aligned} \quad (91)$$

Combining equations 91 and 90 yields Maxwell-Faraday's law [2]:

$$\nabla \times \mathbf{E} = \frac{4\pi k_e q}{|\mathbf{r}|^3} \quad \wedge \quad \frac{\partial \mathbf{B}}{\partial t} = -\frac{4\pi k_e q}{|\mathbf{r}|^3} \Rightarrow \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}. \quad (92)$$

Maxwell's wave equations

From the Gauss's electric field equations 81:

$$\mathbf{E} = \frac{2\pi k_e q}{|\mathbf{r}|^2} \Rightarrow \nabla^2 \mathbf{E} = \frac{12\pi k_e q}{|\mathbf{r}|^4}. \quad (93)$$

$$\mathbf{E} = \frac{2\pi k_e q}{|\mathbf{r}|^2} \wedge r = ct \Rightarrow \mathbf{E} = \frac{2\pi k_e q}{(ct)^2} \Rightarrow \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4}. \quad (94)$$

$$\frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4} \wedge r = ct \Rightarrow \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\mathbf{r}|^4}. \quad (95)$$

$$\nabla^2 \mathbf{E} = \frac{12\pi k_e q}{|\mathbf{r}|^4} \wedge \frac{\partial^2 \mathbf{E}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\mathbf{r}|^4} \Rightarrow \nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}. \quad (96)$$

$$\nabla^2 \mathbf{E} = \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} \wedge \varepsilon_0 \mu_0 = \frac{1}{4\pi k_e} \cdot \frac{4\pi k_e}{c^2} = \frac{1}{c^2} \Rightarrow \nabla^2 \mathbf{E} = \varepsilon_0 \mu_0 \frac{\partial^2 \mathbf{E}}{\partial t^2}, \quad (97)$$

which is Maxwell's electric field wave equation [2]. And from equation (90):

$$\frac{\partial \mathbf{B}}{\partial t} = -\frac{4\pi k_e q}{r^3} \wedge r = |\mathbf{r}| \Rightarrow \nabla^2 \mathbf{B} = \frac{12\pi k_e q}{|\mathbf{r}|^4}. \quad (98)$$

$$\mathbf{B} = \frac{2\pi k_e q}{|\mathbf{r}|^2} \wedge r = ct \Rightarrow \mathbf{B} = \frac{2\pi k_e q}{(ct)^2} \Rightarrow \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4}. \quad (99)$$

$$\frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{(12\pi k_e/c^2)q}{t^4} \wedge r = ct \Rightarrow \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\mathbf{r}|^4}. \quad (100)$$

$$\nabla^2 \mathbf{B} = -\frac{12\pi k_e q}{|\mathbf{r}|^4} \wedge \frac{\partial^2 \mathbf{B}}{\partial t^2} = \frac{12\pi k_e c^2 q}{|\mathbf{r}|^4} \Rightarrow \nabla^2 \mathbf{B} = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2}. \quad (101)$$

$$\nabla^2 \mathbf{B} = \frac{1}{c^2} \frac{\partial^2 \mathbf{B}}{\partial t^2} \wedge \varepsilon_0 \mu_0 = \frac{1}{4\pi k_e} \cdot \frac{4\pi k_e}{c^2} = \frac{1}{c^2} \Rightarrow \nabla^2 \mathbf{B} = \varepsilon_0 \mu_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}, \quad (102)$$

which is Maxwell's magnetic field wave equation [2].

Classic electron radius, r_e

From the unit ratios 49:

$$r = c_m m \wedge r = c_q q \Rightarrow c_m m = c_q q \Rightarrow m = (c_q/c_m)q. \quad (103)$$

$$m = (c_q/c_m)q \wedge r = c_q q \Rightarrow mr = (c_q^2/c_m)q^2 = \frac{r_p m_p q^2}{q_p^2}. \quad (104)$$

$$m_e r_e = \frac{r_p m_p q_e^2}{q_p^2} \Rightarrow r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} \approx 2.8179402276 \cdot 10^{-15} m, \quad (105)$$

which is classic electron radius value [17], where m_e is the electron mass, q_e is the electron charge, and r_p , m_p , and q_p are the Planck length, mass, and charge units.

Fine structure constants, α and α_G

Parsimonious explanations for the fine structure electron charge coupling constant, α , [17] and the fine structure electron gravity coupling constant, α_G , are the ratio of two subtypes of a force. Where m_e is the electron mass, q_e is the electron charge, and r_p , m_p , and q_p are the Planck length, mass, and charge units:

$$\frac{F_{q_e}}{F_{q_p}} = \frac{k_e q_e^2 / r^2}{k_e q_p^2 / r^2} = \frac{q_e^2}{q_p^2} \approx 0.0072973525643 = \alpha. \quad (106)$$

$$\frac{F_{m_e}}{F_{m_p}} = \frac{G m_e^2 / r^2}{G m_p^2 / r^2} = \frac{m_e^2}{m_p^2} \approx 1.751809 \cdot 10^{-45} = \alpha_G, \quad (107)$$

Rearranging the classic electron radius equation (105) also yields α and α_G :

$$r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} \Rightarrow \frac{r_e m_e}{r_p m_p} = \frac{q_e^2}{q_p^2} = \alpha. \quad (108)$$

$$r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} \Rightarrow \left(\frac{r_p q_e^2}{r_e q_p^2} \right)^2 = \frac{m_e^2}{m_p^2} = \alpha_G. \quad (109)$$

Rydberg constant, R_∞

The wave number, R_∞ , is a multiple of $1/r_p$, where the multiple is a function of the ratios of the gravity and charge forces, α_G and α , and the ratio of the classic electron and Planck radii, r_e and r_p . Starting with the ratio-derived classic electron radius (105):

$$\begin{aligned} r_e = r_p \frac{m_p q_e^2}{m_e q_p^2} &\Rightarrow \frac{r_e}{r_p} = \frac{m_p q_e^2}{m_e q_p^2} \Rightarrow \frac{r_e}{r_p} \left(\frac{\alpha_G \alpha}{4\pi r_p} \right) = \frac{m_p q_e^2}{m_e q_p^2} \left(\frac{\alpha_G \alpha}{4\pi r_p} \right) \\ &\Rightarrow \frac{r_e}{4\pi r_p^2} \frac{m_e^2 q_e^2}{m_p^2 q_p^2} = \frac{m_e q_e^4}{4\pi r_p m_p q_p^4} \approx 1.09737315 \cdot 10^7 \text{ m}^{-1} = R_\infty, \end{aligned} \quad (110)$$

where m_e is the electron mass, q_e is the electron charge, and r_p , m_p , and q_p are Planck units.

$\hbar$, h , and Planck-Einstein relation

Applying both the direct proportion ratio (51), and inverse proportion ratio (49):

$$m = k_m / r \quad \wedge \quad r = ct \quad \Rightarrow \quad m(ct)^2 = (k_m / r) r^2 = k_m r. \quad (111)$$

$$m(ct)^2 = k_m r \quad \Rightarrow \quad E = mc^2 = k_m r / t^2. \quad (112)$$

$$E = mc^2 = k_m r / t^2 \quad \wedge \quad r / t = c \quad \Rightarrow \quad E = mc^2 = k_m c / t. \quad (113)$$

Solving the fine structure constant equation 108 for $r_p m_p$, where $r_p m_p = k_m$:

$$r_e m_e / r_p m_p = \alpha \approx 0.0072973525705$$

$$\Rightarrow k_m = r_p m_p = r_e m_e / \alpha \approx 3.5176728259 \cdot 10^{-43} \text{ m} \cdot \text{kg}. \quad (114)$$

$$c_t = c = 2.99792458 \cdot 10^8 \text{ m} \cdot \text{s}^{-1} \quad \wedge \quad k_m \approx 3.5176728259 \cdot 10^{-43} \text{ m} \cdot \text{kg}$$

$$\Rightarrow k_m c \approx 1.0545717829 \cdot 10^{-34} \text{ k} \cdot \text{m}^2 \cdot \text{s}^{-1} = \hbar. \quad (115)$$

$$E = mc^2 = k_m c / t \quad \wedge \quad k_m c = \hbar \quad \wedge \quad f = 1/t \quad \Rightarrow \quad E = mc^2 = \hbar / t = \hbar f. \quad (116)$$

$$E = mc^2 = \hbar / t \quad \wedge \quad \omega \stackrel{\text{def}}{=} 2\pi / t \quad \wedge \quad f = 1/t \quad \wedge \quad h \stackrel{\text{def}}{=} 2\pi \hbar$$

$$\Rightarrow 2\pi E = 2\pi mc^2 = \hbar(2\pi / t) = \hbar \omega = (2\pi \hbar) / t = hf. \quad (117)$$

Values of the unit ratios and Planck units

Values of the direct proportion ratios, c_t , c_m , and c_q are:

$$c_t = c = 2.99792458 \cdot 10^8 \text{ m} \cdot \text{s}^{-1} \text{ (the speed of light)}. \quad (118)$$

$$G = c_m c_t^2 \quad \wedge \quad G \approx 6.67430(15) \cdot 10^{-11} \text{ m}^3 \cdot \text{kg}^{-1} \cdot \text{s}^{-2} \quad \Rightarrow$$

$$c_m = G / c_t^2 \approx 7.426162 \cdot 10^{-28} \text{ m} \cdot \text{kg}^{-1}. \quad (119)$$

$$k_e = c_q^2 c_t^2 / c_m \quad \wedge \quad k_e \approx 8.9875517923(13) \cdot 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2} \quad \Rightarrow$$

$$c_q = \sqrt{k_e c_m / c_t^2} \approx 8.617518 \cdot 10^{-18} \text{ m} \cdot \text{C}^{-1}. \quad (120)$$

From equation 114, the values of the inverse proportion ratios, k_t , k_m , and k_q are:

$$k_m = r_p m_p = r_e m_e / \alpha \approx 3.5176728259 \cdot 10^{-43} \text{ m} \cdot \text{kg}. \quad (121)$$

$$r_p = \sqrt{r_p^2} = \sqrt{c_m k_m} \approx 1.616255 \cdot 10^{-35} \text{ m}. \quad \Rightarrow \quad (122)$$

$$k_t = r_p^2 / c_t \approx 8.71363 \cdot 10^{-79} \text{ m} \cdot \text{s}. \quad (123)$$

$$k_q = r_p^2 / c_q \approx 3.031361 \cdot 10^{-53} \text{ m} \cdot \text{C}. \quad (124)$$

Values of the Planck units, r_p , t_p , m_p , and q_p are:

$$r_p = \sqrt{c_t k_t} = \sqrt{c_m k_m} = \sqrt{c_q k_q} \approx 1.616255 \cdot 10^{-35} \text{ m}. \quad (125)$$

$$t_p = r_p / c_t \approx 5.391246 \cdot 10^{-44} \text{ s}. \quad (126)$$

$$m_p = r_p / c_m \approx 2.176434 \cdot 10^{-8} \text{ kg}. \quad (127)$$

$$q_p = r_p / c_q = q_e / \sqrt{\alpha} \approx 1.8755460384 \cdot 10^{-18} \text{ C}. \quad (128)$$

Compton wavelength, λ

From equations 51 and 115, where the wavelength $\lambda = 2\pi r$ [2, 18]:

$$r = k_m/m \quad \wedge \quad k_m c = \hbar \quad \Rightarrow \quad \lambda = 2\pi r = \frac{2\pi k_m}{m} = \frac{2\pi k_m}{m} \frac{c}{c} = \frac{2\pi \hbar}{mc} = \frac{h}{mc}. \quad (129)$$

Schrödinger's wave equation

Start with the previous unit ratio-derived reduced Planck-Einstein relation 116, where $V(r, t)$ is the potential energy in a volume. And multiply the kinetic energy component by mc/mc :

$$\forall V(r, t) : V(r, t) = E/2 \quad \wedge \quad E = mc^2 = \hbar/t \quad \Rightarrow \quad \hbar/t = \hbar/2t + V(r, t). \quad (130)$$

$$\hbar/t = \hbar/2t + V(r, t) \quad \Rightarrow \quad \hbar/t = (\hbar/2t)(mc/mc) + V(r, t). \quad (131)$$

use the Use the ratio, $r = ct$ (49):

$$\hbar/t = \hbar mc/2mct + V(r, t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar mc^2/2mcr + V(r, t). \quad (132)$$

$$\hbar/t = \hbar mc^2/2mcr + V(r, t) \quad \wedge \quad E = mc^2 = \hbar/t \quad \Rightarrow \quad \hbar/t = \hbar^2/2mcr + V(r, t). \quad (133)$$

$$\hbar/t = \hbar^2/2mcr + V(r, t) \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/t = \hbar^2/2mr^2 + V(r, t). \quad (134)$$

$$\begin{aligned} \forall m, n, k, \omega \in \mathbb{R} : k = 1/nr_p = 1/r \quad \wedge \quad \omega = 1/mt_p = 1/t \\ \Rightarrow \quad \hbar\omega = \hbar^2 k^2/2m + V(r, t). \end{aligned} \quad (135)$$

Multiply both sides of equation 135 by a wave function, $\Psi(r, t)$:

$$\hbar\omega = \hbar^2 k^2/2m + V(r, t) \quad \Rightarrow \quad \hbar\omega\Psi(r, t) = (\hbar^2 k^2/2m)\Psi(r, t) + V(r, t)\Psi(r, t). \quad (136)$$

$$\begin{aligned} \Psi(r, t) = e^{i(kr - \omega t)} \quad \Leftrightarrow \quad \partial\Psi(r, t)/\partial t = -i\omega\Psi(r, t) \quad \wedge \quad \partial^2\Psi(r, t)/\partial r^2 = -k^2\Psi(r, t) \\ \wedge \quad \hbar\omega\Psi(r, t) = (\hbar^2 k^2/2m)\Psi(r, t) + V(r, t)\Psi(r, t) \\ \Rightarrow \quad \hbar\partial\Psi(r, t)/\partial t = -(\hbar^2/2m)\partial^2\Psi(r, t)/\partial r^2 + V(r, t)\Psi(r, t), \end{aligned} \quad (137)$$

which is the one-dimensional position-space Schrödinger's equation [5, 18].

Dirac's wave equation

Start with the previous unit ratio-derived reduced Planck-Einstein relation 116:

$$\hbar/t = mc^2 \Leftrightarrow \hbar/2t + \hbar/2t = mc^2 \Leftrightarrow \hbar/2t = -\hbar/2t + mc^2. \quad (138)$$

$$\hbar/2t = -\hbar/2t + mc^2 \quad \wedge \quad r = ct \quad \Rightarrow \quad \hbar/2t = -\hbar c/2r + mc^2. \quad (139)$$

$$\begin{aligned} \forall m, n, k, \omega \in \mathbb{R} : k = 1/nr_p = 1/2r \quad \wedge \quad \omega = 1/mt_p = 1/2t \\ \Rightarrow \quad \hbar\omega = -\hbar ck + mc^2. \end{aligned} \quad (140)$$

$$\hbar\omega = -\hbar ck + mc^2 \quad \Rightarrow \quad \hbar\omega\psi = -\hbar ck\psi + mc^2\psi. \quad (141)$$

$$\psi = e^{i(kr - \omega t)} \Leftrightarrow \partial\psi/\partial t = -i\omega\psi \quad \wedge \quad \nabla\psi = ik\psi$$

$$\wedge \quad \hbar\omega\psi = -\hbar ck\psi + mc^2\psi \quad \Rightarrow \quad i\hbar\partial\psi/\partial t = -i\hbar c\nabla\psi + mc^2\psi. \quad (142)$$

$$i\hbar\partial\psi/\partial t = -i\hbar c\nabla\psi + mc^2\psi$$

$$\Rightarrow \quad \exists \alpha, \beta : i\hbar\partial\psi/\partial t = i\hbar c\alpha \cdot -\nabla\psi + mc^2\beta\psi, \quad (143)$$

which is the relativistic, free particle Dirac equation, where $\beta = \gamma^0$, $\alpha = \gamma^0\gamma^k$, $k \in \{1, 2, 3\}$, and each γ a matrix [16].

Total of a type

Applying both the direct (49) and inverse proportion ratios (51) to the Euclidean distance:

$$\begin{aligned} r = \sqrt{r_1^2 + r_2^2}, \quad r_1 = c_\tau\tau, \quad r_2 = k_\tau/\tau, \quad \tau \in \{t, m, q\} \\ \Rightarrow \quad r = \sqrt{(c_\tau\tau)^2 + (k_\tau/\tau)^2} \quad \Leftrightarrow \quad \tau = \sqrt{(r/c_\tau)^2 + (k_\tau/r)^2}. \end{aligned} \quad (144)$$

Quantum extension to Schwarzschild's metric

The simplest way to demonstrate how to add quantum physics to general relativity is by extending Schwarzschild's gravitational time dilation equation and black hole metric. Apply the total of a type equation 144 to the derivation of Schwarzschild's time dilation and metric (64):

$$\begin{aligned} m/\sqrt{(r/c_m)^2 + (k_m/r)^2} = 1 \quad \wedge \quad \sqrt{1 - (v^2/c^2)} = \sqrt{1 - (1)(v^2/c^2)} \\ \Rightarrow \quad \sqrt{1 - (v^2/c^2)} = \sqrt{1 - (m/\sqrt{(r/c_m)^2 + (k_m/r)^2})(v^2/c^2)} \end{aligned} \quad (145)$$

$$\begin{aligned} \sqrt{1 - (v^2/c^2)} = \sqrt{1 - (mv^2/\sqrt{(r/c_m)^2 + (k_m/r)^2}c^2)} \quad \wedge \quad KE = mv^2/2 = mv_{escape}^2 \\ \Rightarrow \quad \sqrt{1 - (v^2/c^2)} = \sqrt{1 - (2mv_{escape}^2/\sqrt{(r/c_m)^2 + (k_m/r)^2}c^2)} \end{aligned} \quad (146)$$

For a photon, the escape velocity, $v_{\text{escape}} = c$.

$$\begin{aligned}\sqrt{1 - (v^2/c^2)} &= \sqrt{1 - (2mv_{\text{escape}}^2/\sqrt{(r/c_m)^2 + (k_m/r)^2}c^2)} \wedge v_{\text{escape}} = c \\ \Rightarrow \sqrt{1 - (v^2/c^2)} &= \sqrt{1 - (2mc^2/\sqrt{(r/c_m)^2 + (k_m/r)^2}c^2)}\end{aligned}\quad (147)$$

Schwarzschild defined the black hole event horizon radius, $\alpha \stackrel{\text{def}}{=} 2Gm/c^2$ [14, 15].. The quantum-extended Schwarzschild radius is $r_s \stackrel{\text{def}}{=} 2m$.

$$\sqrt{1 - (v^2/c^2)} = \sqrt{1 - (2mc^2/\sqrt{(r/c_m)^2 + (k_m/r)^2}c^2)} \wedge r_s \stackrel{\text{def}}{=} 2m \quad (148)$$

$$\Rightarrow \sqrt{1 - (v^2/c^2)} = \sqrt{1 - (r_s/\sqrt{(r/c_m)^2 + (k_m/r)^2})} \quad (149)$$

$$\begin{aligned}r' = r\sqrt{1 - (v^2/c^2)} \wedge \sqrt{1 - (v^2/c^2)} &= \sqrt{1 - (r_s/\sqrt{(r/c_m)^2 + (k_m/r)^2})} \\ \Rightarrow r' &= r\sqrt{1 - (r_s/\sqrt{(r/c_m)^2 + (k_m/r)^2})}\end{aligned}\quad (150)$$

At this point, taking the same steps as taken in equations 72 through 73 yields:

$$\begin{aligned}g_{\mu,\nu} &= \text{diag}[(1 - (r_s/\sqrt{(r/c_m)^2 + (k_m/r)^2})), -(1 - (r_s/\sqrt{(r/c_m)^2 + (k_m/r)^2}))^{-1}, \\ &\quad -((r/c_m)^2 + (k_m/r)^2), -((r/c_m)^2 + (k_m/r)^2)\sin^2\theta].\end{aligned}\quad (151)$$

Quantum extension to Newton's gravity force

The quantum mass effect is easier to understand in the context Newton's gravity equation than in general relativity, because the metric equations and solutions in the EFEs are much more complex. From equations 144 and 49:

$$\begin{aligned}m &= \sqrt{(r/c_m)^2 + (k_m/r)^2} \wedge t = \sqrt{(r/c_t)^2 + (k_t/r)^2} \wedge r = \sqrt{(c_m m)^2 + (k_m/m)^2} \\ \Rightarrow F &= mr/t^2 = m\sqrt{(c_m m)^2 + (k_m/m)^2}/((r/c_t)^2 + (k_t/r)^2).\end{aligned}\quad (152)$$

Quantum extension to Coulomb's charge force

$$\begin{aligned}r &= \sqrt{(c_q q)^2 + (k_q/q)^2} \wedge t = \sqrt{(r/c_t)^2 + (k_t/r)^2} \\ \Rightarrow r/t^2 &= \sqrt{(c_q q)^2 + (k_q/q)^2}/((r/c_t)^2 + (k_t/r)^2).\end{aligned}\quad (153)$$

$$\Rightarrow F = mr/t^2 = m\sqrt{(c_q q)^2 + (k_q/q)^2}/((r/c_t)^2 + (k_t/r)^2).\quad (154)$$

INSIGHTS AND IMPLICATIONS

1. The volume and distance equations were derived from set and number properties without relying on the geometric primitives and relations in Euclidean geometry [12, 13], axiomatic geometry [19–23], trigonometry [24, 25], calculus [24, 26, 27], and vector analysis [4],
2. In this article, all volume-spanning distance measures are an inverse (bijective) function of a volume, v , where v is the sum of subset volumes (Euclidean or non-Euclidean), $g(s_{i,1}, \dots, s_{i,n})$. Notionally:

$$\forall n \in \mathbb{N}, \exists d, v, f, g \in \mathbb{R} : v = f(d), d = f^{-1}(v) = f^{-1}(\sum_{i=1}^m g(s_{i,1}, \dots, s_{i,n})). \quad (155)$$

3. Area-spanning distance as an inverse function of an area, A , where that area is the sum (and subtraction, where a_i, b_i is $\pm$ -signed) of areas (16) is an easy to understand motivation of the dot product:

$$d^2 = A = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}. \quad (156)$$

In contrast, the common justifications for defining the dot product requires pointing to diagrams, trigonometry, matrix operations, and inventing more definitions having no geometric motivation, like Kronecker's delta function, δ_{ij} , where:

$$\sqrt{\sum_{i=1}^n a_i^2} \sqrt{\sum_{i=1}^n b_i^2} \cos(\theta) \stackrel{\text{def}}{=} \|\mathbf{a}\| \|\mathbf{b}\| \cos(\theta) \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}, \quad \text{and} \quad (157)$$

$$\begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \cdot \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix} \stackrel{\text{def}}{=} \sum_{i=1}^n a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}, \quad \text{and} \quad (158)$$

$$d^2 = \sum_{i=1}^m a_i \delta_{ij} b_j = \sum_{i=1}^m a_i b_i \stackrel{\text{def}}{=} \mathbf{a} \cdot \mathbf{b}, \quad \delta_{ij} \stackrel{\text{def}}{=} \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases} \quad (159)$$

4. The left side of the distance sum inequality (.7),

$$(\sum_{i=1}^m (a_i^n + b_i^n))^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}, \quad (160)$$

differs from the left side of Minkowski's sum inequality [11].

$$(\sum_{i=1}^m (a_i^n + b_i^n)^{\mathbf{n}})^{1/n} \leq (\sum_{i=1}^m a_i^n)^{1/n} + (\sum_{i=1}^m b_i^n)^{1/n}. \quad (161)$$

- (a) The two inequalities are the same only where $n = 1$.
 - (b) The distance sum inequality (.7) is a more fundamental inequality because the proof does not require the convexity and Hölder's inequality assumptions required to prove the Minkowski sum inequality [11].
 - (c) $\forall n > 1, v > 0$: The left side of the distance sum inequality, $v_i^n = a_i^n + b_i^n$: $\Rightarrow d = v^{1/n} = (\sum_{i=1}^m v_i^n)^{1/n}$, is the Minkowski distance, which makes the distance sum inequality directly related to geometry. But the left side of the Minkowski sum inequality, $d = v^{1/n} = (\sum_{i=1}^m ((v_i^n)^{\mathbf{n}}))^{1/n} = (\sum_{i=1}^m v_i^{\mathbf{n}^2})^{1/n}$, is *not* a Minkowski distance – not as directly related to geometry.
 - (d) The distance sum inequality might be useful for comparing distances between concepts in machine learning.
5. *Combinatorics*: The number of ordered combinations (n-tuples) of the elements of disjoint sets was proven to imply the Euclidean volume equation (.1). And the notion of real-valued, volume spanning distance was derived from countable distance as the inverse function of the number of n-tuples.
 6. *Combinatorics*: The combinatorial AACL property (.16) implies at most 3 volume dimensions and 3 non-volume dimensions, where $\subset \mathbb{R}$ is the common property.
 - (a) If there are more than 3 volume and 3 non-volume dimensions $\subseteq \mathbb{R}$, then the additional dimensions are either not commutative or not linear sequenceable.
 - (b) There cannot be an AACL of more than 3 quantum states of the same type, for example, 3 charge colors and 3 anti-charge colors.
 7. *Combinatorics*: The combinatorial AACL property (.16) implies that for each constant unit interval length, r_p , of an Euclidean distance, there are only 3 possible correspondences to non-distance constant unit interval lengths: 3 direct constant proportion unit ratios (49): $r = (r_p/t_p)t = (r_p/m_p)m = (r_p/q_p)q$ and 3 inverse constant proportion ratios (51): $r = (r_p t_p)/t = (r_p m_p)/m = (r_p q_p)/q$, where r_p , t_p , m_p , and q_p are the Planck units (125) [2, 5].
 8. *Combinatorics*: In biology, there are only 4 deoxyribose nucleotides (DNA), where different sequences of combinations of DNA, yield all the characteristics of millions of

unimaginably complex forms of life. Reductionism would suggest a couple or a few set and number properties or few space, time, etc. properties, where different sequences of combinations, yield all types of particles.

9. A discrete (point) valued state has measure 0 (zero-length interval). The ratio of a time, distance, mass, or charge interval length to zero is undefined, which is why discrete, entangled state values change independent of distance and time.
10. In this article, the following constants are derived directly from the ratios, $G = c_m c_t^2$ (54), charge, $k_e = c_q^2 c_t^2 / c_m$ (77), permittivity $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2 / c_m)$ (81), and permeability $\mu_0 = 4\pi k_e / c_t^2 = 4\pi c_q^2 / c_m$ (85), reduced Planck $\hbar = k_m c_t$ (116).

And the quantum extensions to: Schwarzschild's gravitational time dilation (147), Newton's gravity force (152), and Coulomb's charge force (154), show that the constants G , $\kappa = 8\pi G/c^4$, k_e , and ε_0 do not exist, are decomposed, where the quantum effects are measurable.

Therefore, the constants: G , κ , k_e , ε_0 , and μ_0 are *not* fundamental constants because 1) they were all shown to be composed of Planck unit ratios and 2) G , κ , k_e , and ε_0 do not exist, are decomposed, where the quantum effects are measurable.

11. c_t ($c_t = c$, the speed of light) is a component of the constants: $G = c_m c_t^2$, $\kappa = 8\pi G/c^4 = 8\pi c_m/c^2$, $k_e = c_q^2 c_t^2 / c_m$, $\varepsilon_0 = 1/4\pi k_e = 1/4\pi(c_q^2 c_t^2 / c_m)$, and $\hbar = k_m c_t$. Therefore, the common practice in relativity and particle physics of setting $c = 1$ in various equations should only be done if G , κ , k_e , ε_0 , $\hbar$ and h are also normalized (set to 1).
12. The constants, derived in this article, that do not contain c_t are: vacuum permeability: $\mu_0 = 4\pi k_e / c_t^2 = 4\pi c_q^2 / c_m$, the classic electron radius, $r_e = r_p m_p q_e^2 / m_e q_p^2$, the fine structures, $\alpha = q_e^2 / q_p^2$, and $\alpha_G = m_e^2 / m_p^2$, and Rydberg, $R_\infty = m_e q_e^4 / 4\pi r_p m_p q_p^4$.
13. The magnetic field, $\mathbf{B}$, and magnetic permeability, μ_0 , are derived in the process of deriving Lorentz's law (87). $\mathbf{B} = (2\pi k_e / c^2) v q / |\mathbf{r}|^2$, and where $v = c$, allows a simple derivations of the Faraday's law (92) and the magnetic component of Maxwell's electromagnetic equations (102).
14. Substituting: mass, m , for charge, q , the gravity constant, G for the charge constant k_e , the gravity field, $\mathbf{g}$ for the electric field, $\mathbf{E}$, a time field, $\mathbf{A}$, for the magnetic field,

$\mathbf{B}$, in the Lorentz law derivation steps 82 through 87, yields a new law:

$$\mathbf{F} = m(\mathbf{g} - \mathbf{v} \times \mathbf{A}). \quad (162)$$

The field, $\mathbf{A} = (-2\pi G/c^2)vm/|\mathbf{r}|^2$, has the SI units, s^{-1} , a time measure perpendicular to the gravity field, where time moves in inverse proportion to the magnitude of $\mathbf{g}$.

The fields $\mathbf{g}$ and $\mathbf{A}$ also suggest gravity-time laws analogous to Faraday's law and Maxwell's wave equations.

15. Other derivations of Maxwell's electric and magnetic field wave equations require: Gauss's electric field law in a vacuum: $\nabla \cdot \mathbf{E} = -\rho/\varepsilon_0$, Gauss's magnetic field law in a vacuum: $\nabla \cdot \mathbf{B} = 0$, Faraday's Law: $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$, and Ampere's Law: $\nabla \times \mathbf{B} = \mu_0\varepsilon_0\partial\mathbf{E}/\partial t$ [2]. In contrast, the derivations (97) (102), in this article, only used the speed of light ratio, $r = ct$ and the ratio-derived $\mathbf{E}$ and $\mathbf{B}$ equations.
16. As shown in the derivation of the Schwarzschild's black hole metric 73, where the laws of physics are derived as ratios, the metric, $g_{\mu,\nu}$, can be derived independent of Einstein's field equation (EFE) and derived without the complexity of manipulating Christoffel symbols, using determinants, etc. [14, 15].
17. In this article, quantum extensions to general relativity were implemented by extending the Schwarzschild metric (151) rather than trying extend the EFE (61). Likewise, it would be more rigorous to extend the metric, $g_{\mu,\nu}$, using special relativity and the ratios to better match cosmological observation than to add a fudge factor, $\Lambda g_{\mu,\nu}$, to the EFE (61).
18. c_t , c_m , and c_q are the *maximum* proportionality constants: $r = \sqrt{r_1^2 + r_2^2 + r_3^2} \Rightarrow r_1, r_2, r_3 \leq r \Rightarrow \forall \tau \in \{t, m, q\} : r_1/\tau, r_2/\tau, r_3/\tau \leq r/\tau = r_p/\tau_p = c_\tau$. For example, the speeds of gravity and light waves are limited to the proportionality constant, c_t .
19. Einstein's relativity equations assume: 1) the Lorentz transformations, 2) the laws of physics are same at each local Euclidean-like coordinate point, 3) the notion of light, and 4) the speed of light is the same at each coordinate point [3, 4].

The derivations, in this article, were made without those assumptions (does not even require the notion of light). The unit ratios are *constant* ratios of a *constant* Euclidean distance interval length to *constant* interval lengths of time, mass, and charge.

Therefore, the *constant* unit ratios and all equations (laws) and constants derived from those ratios and Euclidean distance are the same at each local Euclidean-like coordinate point.

20. Some common assumptions about the reduced and full Planck constants are incorrect!

- (a) The reduced Planck constant, $\hbar$, was shown to be composed of the proportionality constants, k_m and c_t , where: $k_m c_t = \hbar$ (114). Therefore, the common assumption that the Planck constant is not derived from other constants is *incorrect*.
- (b) The rigorously derived reduced Planck-Einstein equation, $E = mc^2 = \hbar/t = \hbar f$, (116) implies $2\pi E = 2\pi mc^2 = (2\pi\hbar)f = hf$ (117). Therefore, the assumed Planck-Einstein relation, $E = mc^2 = hf$, is incorrect by a factor of 2π .

Further, the derivations, in this article, of Schrödinger's wave equation (137) [5] and Dirac's wave equation (143) [16] from the ratio-derived, reduced Planck-Einstein relation, $E = mc^2 = \hbar/t$ (116), are more rigorous (do not require the De Broglie wavelength assumption [2]).

Therefore, it would be logical for CODATA to define the reduced Planck constant as $\hbar = k_m c_t$ and define the full Planck constant, h , as, $h = 2\pi\hbar$.

21. An empirical law and related constant *describes* the relations between the variables in an equation, where the units (dimensions) of the constant are added ad hoc to make the units on both sides of an equation balance. Dimensional analysis is used to define the constant in terms of other constants, for example, the CODATA definition of the fine structure constant, $\alpha = q_e^2/4\pi\epsilon_0\hbar c$ [17]. The constant definition resulting from dimensional analysis provides *no* understanding of the physics principles that causes the definition, thereby, making the constant definition “opaque.”

In contrast, where an equation is derived from the ratios, the ratios are the *reason/explanation* for the relations between the variables. Further, the ratios have units (dimensions). Therefore, all equations and constants derived from the ratios have dimensions, which eliminates the need for dimensional analysis.

- (a) The opaque CODATA definition, of the Planck length constant, $r_p = \sqrt{\hbar G/c^3}$ [17], derived from dimensional analysis, simplifies to the ratio-derived equation,

in this article (125): $r_p = \sqrt{\hbar G/c^3} = \sqrt{(k_m c)(c_m c^2)/c^3} = \sqrt{k_m c_m}$.

- (b) The opaque CODATA definition, $\alpha = q_e^2/4\pi\epsilon_0\hbar c$, [17], derived from dimensional analysis, reduces to the simpler, ratio-derived equation, in this article:

$$\begin{aligned}\epsilon_0 &= 1/4\pi k_e = 1/(4\pi(c_q^2 c_t^2/c_m)) \quad \wedge \quad \hbar = k_m c_t \quad \wedge \quad h = 2\pi\hbar \\ \Rightarrow \quad \epsilon_0 \hbar c &= 2\pi k_m c_t^2 / (4\pi(c_q^2/c_m)c_t^2) = k_m / (2(c_q^2/c_m)) \\ &= m_p r_p / (2((r_p/q_p)^2/(r_p/m_p))) = q_p^2/2. \quad (163)\end{aligned}$$

$$\alpha = q_e^2/2\epsilon_0 \hbar c \quad \wedge \quad \epsilon_0 \hbar c = q_p^2/2 \quad \Rightarrow \quad \alpha = q_e^2/2(q_p^2/2) = q_e^2/q_p^2. \quad (164)$$

The explanation for the ratio-based definition is the ratio of two subtypes of Coulomb's charge force: $\alpha = q_e^2/q_p^2 = F_{q_e}/F_{q_p}$ (106).

- (c) The opaque fine structure electron gravity coupling constant, $\alpha_G = Gm_e^2/\hbar c$, derived from dimensional analysis, reduces to the simpler, ratio-derived equation:

$$\alpha_G = Gm_e^2/\hbar c = (c_m c_t^2)m_e^2/(k_m c_t)c_t = (r_p/m_p)m_e^2/m_p r_p = m_e^2/m_p^2. \quad (165)$$

The explanation for the ratio-based definition is the ratio of two subtypes of Newton's gravity force: $\alpha_G = m_e^2/m_p^2 = F_{m_e}/F_{m_p}$ (107).

- (d) The opaque NIST/CODATA definition of the Rydberg constant [17], derived from dimensional analysis, reduces to the ratio-derived equation (110):

$$\begin{aligned}R_\infty &= \frac{m_e q_e^4}{8\epsilon_0^2 \hbar^3 c} = \frac{m_e q_e^4}{8(1/4\pi k_e)^2 (2\pi\hbar)^3 c_t} = \frac{m_e q_e^4 (4\pi k_e)^2}{8(2\pi\hbar)^3 c_t} = \frac{m_e q_e^4 (c_q^2 c_t^2/c_m)^2}{4\pi (k_m c_t)^3 c_t} \\ &= \frac{m_e q_e^4 (c_q^2/c_m)^2}{4\pi k_m^3} = \frac{m_e q_e^4 c_q^4}{4\pi k_m^3 c_m^2} = \frac{m_e q_e^4 (r_p/q_p)^4}{4\pi (r_p m_p)^3 (r_p/m_p)^2} = \frac{m_e q_e^4}{4\pi r_p m_p q_p^4}. \quad (166)\end{aligned}$$

And from equation 110:

$$R_\infty = \frac{m_e q_e^4}{4\pi r_p m_p q_p^4} = \frac{r_e}{4\pi r_p^2} \frac{m_e^2 q_e^2}{m_p^2 q_p^2} = \frac{1}{r_p} \cdot \frac{r_e}{4\pi r_p} \alpha_G \alpha. \quad (167)$$

The ratio-based explanation is that the wave number, R_∞ , is a multiple of $1/r_p$, where the multiple is a function of the ratios of the gravity and charge forces, α_G and α , and the ratio of the classic electron and Planck radii, r_e and r_p .

- (e) The opaque NIST/CODATA definition of the vacuum permeability constant, $\mu_0 = 2h\alpha/cq_e^2$ [17], derived from dimensional analysis, reduces to a simpler ratio-derived definition. Using the ratios to reduce the old definition of μ_0 derived in

this article:

$$\mu_0 = 4\pi k_e/c_t^2 = 4\pi c_q^2/c_m = 4\pi(r_p/q_p)^2/(r_p/m_p) = 4\pi m_p r_p/q_p^2 = 4\pi k_m/q_p^2. \quad (168)$$

Using the ratios to reduce the CODATA definition, $\mu_0 = 2h\alpha/cq_e^2$ [17]:

$$h = 2\pi k_m c_t, \quad \alpha = q_e^q/q_p^2 \Rightarrow \mu_0 = 2h\alpha/cq_e^2 = 4\pi k_m c_t(q_e^2/q_p^2)/c_t q_e^2 = 4\pi k_m/q_p^2. \quad (169)$$

22. The quantum extensions to: Schwarzschild's black hole metric (151), Newton's gravity force (152), and Coulomb's charge force (154) make quantifiable predictions:

- (a) For both gravity and charge: $\lim_{r \rightarrow 0} F = 0$, where the forces and spacetime curvature peak at the Planck unit length, r_p . Finite maximum gravity and charge forces eliminate the problem of the infinite energy required for forces going to infinity as $r \rightarrow 0$. Further analysis is needed to determine impact on radioactivity, weak force, strong force, and electron tunneling.
- (b) The quantum-extended equations reduce to the non-extended equations, where the distance between 2 masses and between 2 charges is sufficiently large or the masses and charges sufficiently large that the quantum effects are not measurable. *Note* that G , k_e , ε_0 , μ_0 , and $\kappa = 8\pi G/c^4$ do not exist, are decomposed, where the quantum effects becomes measurable.
- (c) The ratio of quantum units, m_p/r_p^3 , might indicate a maximum mass density (where only approximately 10^{19} protons and neutrons can be compressed into a 1 cubic Planck length volume). A finite force (spacetime curvature) and finite mass density might imply that black holes have sizes > 0 (are not singularities), which might impact whether black holes evaporate and whether a big bang would have originated from a singularity.

DATA AVAILABILITY STATEMENT

The supplemental files are computer source code files (ASCII text) expressing formal math proofs used to validate the proofs in this article. The source code files are compiled by a logic engine called Rocq (formerly known as Coq) [10]. If the files compile without error, then the consensus of many mathematicians is that the proofs are likely to be correct.

The supplemental file, “euclidrelations.v,” contains formal proofs about Euclidean volume, dot product, Minkowski distance, distance inequalities, and metric space. The supplemental file, “threed.v,” contains a formal proof of the properties that limit set to at most 3 elements.

-
- [1] I. Newton, *The Mathematical Principles of Natural Philosophy* (Patristic Publishing, 2019) reprint.
 - [2] R. Feynman, R. Leighton, and M. Sands, *The Feynman Lectures on Physics* (California Institute of Technology, 2010).
 - [3] A. Einstein, *Relativity, The Special and General Theory* (Princeton University Press, 2015).
 - [4] H. Weyl, *Space-Time-Matter* (Dover Publications Inc, 1952).
 - [5] E. Schrödinger, *Collected Papers On Wave Mechanics* (Minkowski Institute Press, 2020).
 - [6] R. R. Goldberg, *Methods of Real Analysis* (John Wiley and Sons, 1976).
 - [7] W. Rudin, *Principles of Mathematical Analysis* (McGraw Hill Education, 1976).
 - [8] W. Conradie and V. Goranko, *Logic and Discrete Mathematics* (Wiley, 2015).
 - [9] X. P. N. Guigui, N. Miolane, Introduction to riemannian geometry and geometric statistics: from basic theory to implementation with geomstats, *Foundations and Trends in Machine Learning* **16**, 182 (2023), [hal-00644459](#).
 - [10] Rocq, [Rocq proof assistant](#) (2025).
 - [11] H. Minkowski, *Geometrie der Zahlen* (Chelsea, 1953) reprint.
 - [12] D. E. Joyce, [Euclid’s elements](#) (1998).
 - [13] E. S. Loomis, *The Pythagorean Proposition* (NCTM, 1968).
 - [14] K. Schwarzschild, Über das gravitationsfeld eines massenpunktes nach der einsteinschen theorie, *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften* **7**, 198 (1916).
 - [15] S. Antoci and A. Loinger, An english translation of k. schwarzschild’s 1916 article: On the gravitational field of a mass point according to einstein’s theory, [ARXIV.org](#) (1999).
 - [16] P. Dirac, *The Principles of Quantum Mechanics* (Snowball Publishing, 1957).
 - [17] CODATA, [Physical constant values](#) (2022).
 - [18] M. Jain, *Quantum Mechanics: A Textbook for Undergraduates* (PHI Learning Private Limited, New Delhi, India, 2011).

- [19] G. D. Birkhoff, A set of postulates for plane geometry (based on scale and protractors), *Annals of Mathematics* **33** (1932).
- [20] D. Hilbert, *The Foundations of Geometry (2cd Ed)* (Chicago: Open Court, 1980).
- [21] J. M. Lee, *Axiomatic Geometry* (American Mathematical Society, 2010).
- [22] O. Veblen, A system of axioms for geometry, *Trans. Amer. Math. Soc* **5**, 343 (1904).
- [23] A. Tarski and S. Givant, Tarski's system of geometry, *The Bulletin of Symbolic Logic* **5**, 175 (1999).
- [24] A. Bogomolny, *Pythagorean Theorem* (Interactive Mathematics Miscellany and Puzzles, 2010).
- [25] J. Zimba, On the possibility of trigonometric proofs of the pythagorean theorem, *Forum Geometricorum* **9**, 275 (2009).
- [26] B. C. Berndt, Ramanujan-100 years old (fashioned) or 100 years new (fangled)?, *The Mathematical Intelligencer* **10** (1988).
- [27] M. Staring, The pythagorean proposition a proof by means of calculus, *Mathematics Magazine* **69**, 45 (1996).